MSG No. 97.152 $I4221'$ [Type II, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4221'

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]

Table 2: Wyckoff site: 2b, site symmetry: 4221'

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]

Table 3: Wyckoff site: 4c, site symmetry: 222.1'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	[1,4,5,6,17,20,21,22]
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	[2,3,7,8,18,19,23,24]
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[9,12,13,14,25,28,29,30]
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[10,11,15,16,26,27,31,32]

Table 4: Wyckoff site: 4d, site symmetry: 2.221'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,6,15,16,17,22,31,32]
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[2,3,12,13,18,19,28,29]
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	[4,5,10,11,20,21,26,27]
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[7,8,9,14,23,24,25,30]

Table 5: Wyckoff site: 4e, site symmetry: 4..1'

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2,3,6,17,18,19,22]
2	$[0,\ 0,\ -z]$	[4,5,7,8,20,21,23,24]
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[9,10,11,14,25,26,27,30]
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[12,13,15,16,28,29,31,32]

Table 6: Wyckoff site: 8f, site symmetry: 2..1'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,6,17,22]
2	$\left[\frac{1}{2},\ 0,\ z\right]$	[2,3,18,19]
3	$\left[0,\ \frac{1}{2},\ -z\right]$	[4,5,20,21]

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, -z\right]$	[7,8,23,24]
5	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[9,14,25,30]
6	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[10,11,26,27]
7	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[12,13,28,29]
8	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[15,16,31,32]

Table 7: Wyckoff site: 8g, site symmetry: ..21'

No.	position	mapping
1	$[x, x, 0]$	[1,7,17,23]
2	$[-x, x, 0]$	[2,5,18,21]
3	$[x, -x, 0]$	[3,4,19,20]
4	$[-x, -x, 0]$	[6,8,22,24]
5	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[9,15,25,31]
6	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$	[10,13,26,29]
7	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[11,12,27,28]
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$	[14,16,30,32]

Table 8: Wyckoff site: 8h, site symmetry: .2.1'

No.	position	mapping
1	$[x, 0, 0]$	[1,4,17,20]
2	$[0, x, 0]$	[2,7,18,23]
3	$[0, -x, 0]$	[3,8,19,24]
4	$[-x, 0, 0]$	[5,6,21,22]
5	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9,12,25,28]
6	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[10,15,26,31]
7	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[11,16,27,32]
8	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[13,14,29,30]

Table 9: Wyckoff site: 8i, site symmetry: .2.1'

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,4,17,20]
2	$\left[0, x, \frac{1}{2}\right]$	[2,7,18,23]
3	$\left[0, -x, \frac{1}{2}\right]$	[3,8,19,24]
4	$\left[-x, 0, \frac{1}{2}\right]$	[5,6,21,22]
5	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[9,12,25,28]
6	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[10,15,26,31]
7	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[11,16,27,32]
8	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[13,14,29,30]

Table 10: Wyckoff site: `8j`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	`[1,15,17,31]`
2	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{4}\right]$	`[2,13,18,29]`
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{4}\right]$	`[3,12,19,28]`
4	$\left[x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	`[4,11,20,27]`
5	$\left[-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[5,10,21,26]`
6	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[6,16,22,32]`
7	$\left[x+\frac{1}{2},\, x,\, \frac{3}{4}\right]$	`[7,9,23,25]`
8	$\left[\frac{1}{2}-x,\, -x,\, \frac{3}{4}\right]$	`[8,14,24,30]`

Table 11: Wyckoff site: `16k`, site symmetry: `11'`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1,17]`
2	$[-y,\, x,\, z]$	`[2,18]`
3	$[y,\, -x,\, z]$	`[3,19]`
4	$[x,\, -y,\, -z]$	`[4,20]`
5	$[-x,\, y,\, -z]$	`[5,21]`
6	$[-x,\, -y,\, z]$	`[6,22]`
7	$[y,\, x,\, -z]$	`[7,23]`
8	$[-y,\, -x,\, -z]$	`[8,24]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9,25]`
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[10,26]`
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[11,27]`
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	`[12,28]`
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[13,29]`
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[14,30]`
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[15,31]`
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	`[16,32]`